

BISHOPS HEAD, MD, USAWMO: **997688**Lat: **38.220N**Lon: **76.039W**Elev: **10**StdP: **14.69**Time Zone: **-5.00 (NAE)**Period: **06-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	23.2	N/A	N/A	N/A	N/A	N/A	N/A	32.9	28.8	30.6	27.1	13.5	330	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	88.1	N/A	86.1	N/A	84.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	10.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.8	24.5	21.1	16.3	91.5	2.9	2.6	14.2	93.4	12.5	94.9	10.8	96.4	8.7	98.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	36.1	38.1	45.2	56.2	67.0	75.6	79.5	77.9	71.8	61.7	49.3	41.7
	DBStd	16.59	7.94	7.46	7.86	6.80	6.23	4.92	3.83	3.73	5.02	6.76	6.20	6.45
	HDD50	1336	436	336	189	22	0	0	0	0	4	85	264	
	HDD65	3930	897	753	613	272	52	2	0	6	143	470	722	
	CDD50	4429	4	4	42	208	528	768	916	866	655	367	64	7
	CDD65	1546	0	0	0	8	115	320	451	401	211	40	0	0
	CDH74	11151	0	0	0	7	325	2222	4260	3309	972	56	0	0
	CDH80	2310	0	0	0	0	31	424	1120	640	94	1	0	0
Wind	WSAvg	10.8	10.8	11.1	11.6	11.7	11.5	10.9	10.4	9.7	10.1	11.1	10.7	10.5
Precipitation	PrecAvg	46.0	3.2	2.8	3.9	3.5	3.6	4.4	4.5	5.0	4.2	3.9	3.0	3.7
	PrecMax	65.6	7.2	7.4	10.2	7.8	8.8	10.4	10.3	10.2	10.8	8.0	7.8	8.0
	PrecMin	35.4	1.5	0.5	0.0	1.9	1.1	1.6	1.5	1.5	0.9	0.7	0.4	1.5
	PrecStd	6.9	1.3	1.5	1.9	1.3	2.0	2.1	2.1	2.0	2.6	2.2	1.9	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.2	58.2	65.6	74.8	83.6	88.9	91.4	89.9	84.7	78.4	65.4	60.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	54.1	53.7	62.0	70.8	79.9	86.1	89.0	86.7	82.4	75.4	62.1	56.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	50.7	51.0	59.0	67.8	77.5	84.2	86.9	85.0	80.9	73.5	59.7	53.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	47.0	48.2	56.1	65.3	75.3	82.4	85.2	83.3	79.3	71.0	58.0	50.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	10.5	10.7	9.9	9.1	8.2	8.1	8.5	8.6	9.2	9.0
		MCDBR	12.3	12.8	13.4	11.8	10.5	10.1	9.1	8.4	8.7	9.1	10.4	11.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Clear-Sky Solar Irradiance	taub	0.313	0.320	0.354	0.393	0.437	0.484	0.514	0.506	0.408	0.360	0.327	0.314	
	taud	2.504	2.479	2.425	2.332	2.243	2.162	2.089	2.099	2.350	2.485	2.533	2.540	
	Ebn at Noon	276	289	288	281	269	256	246	245	266	270	269	268	
	Edh at Noon	25	29	34	39	44	48	51	49	36	29	24	23	
All-Sky Solar Radiation	RadAvg	699	959	1284	1628	1847	2022	1957	1755	1425	1085	790	616	
	RadStd	56	100	106	115	198	149	94	101	140	115	68	55	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (75 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+266	+164

Nomenclature: See separate page